



Mini Project Report

on

Predicting Air quality index with contemporary and historical data

Submitted in the partial fulfillment of the requirements

for the degree

Masters in Technology

by

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Abstract

Accurate AQI forecasting demands both high-fidelity data and adaptive modeling. In this project, we ingest hourly IAQI readings for PM_{2.5}, PM₁₀, CO, SO₂, NO₂, and O₃—covering 14,107 cities worldwide—from the AQICN.org API into a MongoDB time-series store via a Google Cloud Run service triggered every hour. After preprocessing (cleaning, interpolation, and breakpoint calculation), we train four complementary frameworks—ARIMA, XGBoost, LSTM, and a discrete-state Markov chain—on each pollutant's historic IAQI. For each pollutant, we automatically select the model with the highest cross-validated accuracy, then compute two parallel AQI forecasts: an Ensemble-AQI, which takes the maximum across pollutant forecasts from their respective best models, and a Dominant-Pollutant AQI, which uses the single highest-accuracy pollutant forecast. Both series, complete with 95% uncertainty bounds, are exposed via RESTful endpoints.

Our end-to-end pipeline—deployed on Google Cloud with hourly scheduling and integrated with a real-time visualization dashboard—enables side-by-side comparison of Ensemble-AQI vs. Dominant-Pollutant AQI plots at country, state, and city levels. Globally, the system achieves an overall AQI R² of 0.909 and up to 99% band-classification accuracy, demonstrating that per-pollutant model selection within a scalable cloud architecture can deliver near-real-time, highly reliable AQI forecasts to support proactive public-health interventions and individual exposure planning.

Key words: AQI, Streamlit, dominantpollutant, Ensemble-AQI

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Chapter 1

Introduction

This chapter introduces the importance of accurate Air Quality Index (AQI) forecasting driven by high-fidelity, real-time data, outlines the objectives and motivations behind the system, defines its scope, and describes the organization of the report. We highlight the use of the AQICN.org API as a rich global data source, the application of diverse machine-learning frameworks—including ARIMA, XGBoost, LSTM, and Markov chains—and the deployment on a scalable Google Cloud + MongoDB architecture with an hourly scheduler.

1.1 Introduction

Air pollution poses a significant risk to public health and is linked to respiratory and cardiovascular diseases, making timely air-quality information critical for both authorities and the general public. The Air Quality Index (AQI) converts complex pollutant measurements into a standardized scale from 0 to 500, facilitating clear communication of air-quality conditions and associated health concerns. However, most existing AQI reporting systems only provide real-time or historical values, lacking reliable forecasts that can inform proactive interventions and policy decisions.

To address this gap, our project harnesses the AQICN.org API, which aggregates real-time air-quality data for over 100 countries and thousands of cities worldwide aqicn.org. We ingest hourly pollutant readings including individual AQI (IAQI) for PM_{2.5}, PM₁₀, CO, SO₂, NO₂, and O₃ into a MongoDB time-series database, leveraging its optimized storage and query performance for temporal data. This rich dataset underpins our multi-model forecasting pipeline, enabling accurate short- and medium-term AQI predictions.

1.2 Objectives

This project aims to develop a fully automated, cloud-native AQI forecasting system that delivers highly accurate short- and medium-term predictions for any location worldwide. By leveraging the extensive, hourly IAQI data provided by AQICN.org and integrating diverse machine-learning frameworks within a scalable Google Cloud infrastructure, we seek to empower stakeholders with actionable, real-time and forecasted air-quality insights.

1. Ingest and Store Global IAQI Data

- Retrieve hourly pollutant data (PM_{2.5}, PM₁₀, CO, SO₂, NO₂, O₃) from AQICN.org for over 14,000 cities using a unique station identifier (idx).
- Store this time-series data in a MongoDB database via an automated Google Cloud Run service triggered hourly using Cloud Scheduler.

2. Preprocess and Standardize Data

- Clean, interpolate, and normalize raw pollutant measurements.
- Compute individual pollutant AQI (IAQI) values using EPA-based linear interpolation methods.

3. Train and Evaluate Multiple Forecasting Models

- Implement four distinct models for each pollutant:
 - **ARIMA** for time-series trend detection
 - **XGBoost** for multivariate regression
 - **LSTM** for deep-learning based sequence prediction
 - **Markov Chain** for categorical AQI band transitions
- Validate model performance using metrics like R², MAE, RMSE, and band classification accuracy.

4. Optimize Pollutant-wise Model Selection

- Automatically select the best-performing model for each pollutant based on cross-validation results.
- Use the selected models to produce accurate individual pollutant forecasts.

5. Compute Dual AQI Predictions

- Generate two AQI outputs:
 - **Best-Model AQI**: using the forecast from the highest-performing single model.
 - **Dominant-Pollutant AQI**: taking the AQI value from the most accurate forecasted pollutant at each time step.
- Analyze and visualize both outputs for performance comparison.

6. Cloud-based Automation and Deployment

- Deploy all components (data ingestion, preprocessing, model inference) on Google Cloud using Cloud Run and Scheduler.
- Maintain configuration and versioning through GitHub.

7. Dashboard for Visualization and Insights

- Serve real-time and forecasted AQI data through RESTful APIs.

- Render AQI plots for both prediction modes (best-model and dominant pollutant) via a dynamic dashboard interface.

8. **Demonstrate High Accuracy and Generalizability**

- Achieve AQI prediction R^2 values above 0.90.
- Ensure pollutant-band classification accuracy $\geq 95\%$, with top performance reaching 99% in Belleville, Ontario, Canada.

1.3 Motivation

Air pollution remains one of the most significant environmental threats to human health, contributing to millions of premature deaths each year. With rapid urbanization and industrialization, especially in developing countries, real-time monitoring and accurate forecasting of air quality have become increasingly important. Timely and reliable AQI forecasts allow governments, health organizations, and individuals to take proactive measures to reduce exposure and mitigate health risks.

Traditional air quality monitoring systems often provide only real-time data, without forecasting capabilities, limiting their usefulness in preventive action. Additionally, many existing systems are restricted to specific regions or suffer from limited data availability. This project is motivated by the need for a globally scalable, data-driven, and automated AQI forecasting system that can serve any city in the world, using real-time data and advanced machine learning techniques.

By leveraging publicly available data from AQICN.org and integrating cloud-based automation, this system aims to democratize access to accurate AQI forecasts. The use of multiple ML models and per-pollutant optimization ensures high reliability and precision, empowering users and authorities alike to make data-informed decisions. Ultimately, this work contributes toward building smarter cities and healthier communities through accessible environmental intelligence.

1.3 Scope

This project is designed to provide a complete, automated, and scalable system for forecasting the Air Quality Index (AQI) across any global location using machine learning techniques. The scope includes the entire pipeline—from data collection and storage to prediction, deployment, and visualization. It emphasizes accuracy, scalability, and real-time applicability to support public health and environmental decision-making.

- **Global Data Collection:**

Fetching hourly pollutant concentration and AQI data for over 14,000 cities using the AQICN.org API, including key pollutants: $PM_{2.5}$, PM_{10} , CO, SO₂, NO₂, and O₃.

- **Cloud-Based Infrastructure:**
Automating data ingestion and model execution using Google Cloud Run and Cloud Scheduler. Data is stored in a MongoDB backend for efficient retrieval and analysis.
- **Data Preprocessing:**
Cleaning and interpolating missing values, standardizing data formats, and computing IAQI values using EPA-compliant breakpoints.
- **Model Development:**
Implementing four machine learning frameworks—ARIMA, XGBoost, LSTM, and Markov Chains—to forecast pollutant-specific AQI values. Each pollutant is assigned its best-performing model based on validation metrics.
- **AQI Computation:**
Generating two AQI outputs per forecast:
 1. **Best-Model AQI** – Based on the pollutant-wise best prediction model.
 2. **Dominant-Pollutant AQI** – Based on the pollutant with the highest impact at a given time.
- **Performance Evaluation:**
Comparing model predictions using R^2 , RMSE, MAE, and classification accuracy. Highest accuracy achieved was 99% classification and 0.909 R^2 in Belleville, Ontario.
- **Visualization:**
Developing a user-friendly dashboard to display real-time and forecasted AQI values at global, national, and local levels.
- **Extensibility:**
Designed to support future enhancements, including additional pollutants, longer forecast horizons, and new machine learning models.

1.4 Organization of the Report

This report is structured into six comprehensive chapters that systematically guide the reader through the development of an intelligent air quality forecasting system. Chapter 1: Introduction sets the stage by explaining the project's background, objectives, motivation, scope, and an overview of the report structure. Chapter 2: Literature Review examines the historical evolution of AQI systems, pollutant-specific impacts, and the role of machine

learning in air quality prediction, while identifying gaps in current research. Chapter 3: Methodology outlines the overall system design, including data acquisition, preprocessing steps, model architectures, performance evaluation metrics, and cloud-based deployment strategies. Chapter 4: System Implementation details the technical execution, such as data pipeline setup, model training via user interface triggers, and real-time visualizations. Chapter 5: Results and Analysis presents the model forecasts, compares predictive performance across pollutants, and identifies the best-suited model for each case using various metrics. Lastly, Chapter 6: Conclusion and Future Scope summarizes the study's outcomes and suggests directions for future enhancements and scalability improvements.

Chapter 2

Literature Review

Air pollution forecasting has become an increasingly important field of research due to its direct impact on public health, climate, and urban policy. This chapter reviews the existing literature and methodologies employed in AQI prediction, emphasizing the progression from traditional statistical models to modern machine learning (ML) and deep learning (DL) frameworks, and their integration into cloud-based pipelines for real-time forecasting.

2.1 Origin and Early Developments

Historically, the need to communicate air quality in an understandable format has driven the development of AQI systems globally. These indices have evolved not only as technical tools but also as public health communication instruments [1]. The integration of technological advancements such as machine learning and IoT sensors has further expanded the capabilities of AQI systems, enabling more precise and timely pollution monitoring [2]. Understanding this evolution and its health-driven foundations is essential for appreciating current innovations and identifying areas for improvement.

The concept of AQI emerged from early efforts in the 1960s and 70s to quantify air pollution in urban areas. Initially, indices like the Pollutant Standards Index (PSI) were developed by environmental agencies in the U.S. to report pollutant levels relative to regulatory standards [3]. Over time, the AQI replaced PSI in many countries, offering a more nuanced and public-friendly representation of air quality.

2.1.1 International Standards and Divergence (EPA, CPCB, EU)

Different countries have adopted varied approaches to AQI computation. For instance, the U.S. Environmental Protection Agency (EPA) standardizes AQI based on six major pollutants, while India's Central Pollution Control Board (CPCB) uses a slightly modified version with regional pollutant thresholds [4]. The European Union (EU) uses the Common Air Quality Index (CAQI), which varies in scale and pollutant composition. These differences often stem from variations in pollutant prevalence, local health studies, and regulatory priorities.

2.1.2 Transition from PSI to AQI Systems

The shift from PSI to AQI reflected a broader move toward public transparency and risk-based environmental management. AQI systems integrated health-based breakpoints and started including more pollutants, particularly fine particulate matter (PM_{2.5}), which gained scientific recognition for its harmful effects in the 2000s [3][5].

Changes in Pollutant Consideration and Scaling

The scale and computation of AQI have also evolved. While early indices used fixed thresholds, newer systems apply breakpoint interpolation and statistical methods to estimate AQI more accurately. Technological advancements have enabled automated monitoring stations and centralized databases, improving the responsiveness and coverage of AQI reporting [6].

2.2 Pollutants and Health-Based Breakpoints

2.2.1 Overview of Key Pollutants: PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃

AQI values are typically computed based on concentrations of key pollutants known to adversely affect human health. These include particulate matter (PM_{2.5} and PM₁₀), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃). PM_{2.5} is especially critical due to its ability to penetrate deep into the lungs and bloodstream, causing respiratory and cardiovascular issues [7].

2.2.2 Impact on Human Health and Guideline Values

Each pollutant has specific short- and long-term health implications. For instance, exposure to high NO₂ levels can aggravate asthma, while ozone can trigger chest pain and throat irritation. Health-based breakpoints defined by WHO and national agencies guide the AQI scaling. These breakpoints are determined based on epidemiological studies linking pollutant exposure with adverse health outcomes [1][7].

2.2.3 AQI Computation Methods and Breakpoint Interpolation

AQI calculation methods vary, but most systems use a piecewise linear function (breakpoint interpolation) to convert raw pollutant concentrations into sub-indices. The maximum sub-index across pollutants is then reported as the AQI. Some advanced systems incorporate meteorological factors or time-series predictions using machine learning models like LSTM or Random Forest to enhance AQI forecasting [2][8].

2.3 Global AQI Monitoring Systems

Global AQI monitoring involves collaboration between national and international organizations such as the World Health Organization (WHO), Environmental Protection Agency (EPA) in the U.S., Central Pollution Control Board (CPCB) in India, and AQICN.org, which aggregates global air quality data. These organizations play vital roles

in defining pollutant standards, developing index frameworks, and facilitating cross-border data sharing.

Different countries follow distinct AQI frameworks, leading to divergence in interpretation and scaling. For example, while the U.S. EPA includes six pollutants with a scale up to 500, the Chinese Ministry of Ecology and Environment uses a similar six-pollutant model but with different breakpoints. India's CPCB includes eight pollutants with values often mapped on a 0–500 scale but differs in breakpoint thresholds. The European Union's CAQI uses a 0–100 scale focused more on urban health impacts [1][3].

Data aggregation efforts such as AQICN unify data from over 100 countries, implementing automated quality checks, standardized scaling, and interpolated forecasting using real-time sensor feeds [2][5]. However, challenges persist in harmonizing data formats and managing inconsistencies from low-cost sensor sources.

2.4 Machine Learning for AQI Forecasting

Traditional Models: ARIMA, SARIMA

Time-series forecasting using Auto-Regressive Integrated Moving Average (ARIMA) and its seasonal variant SARIMA has long been used in AQI prediction. These models assume linear temporal dependencies and are effective for univariate pollutant trends but perform poorly with complex, multi-feature interactions [3].

Ensemble and Tree-Based Models: Random Forest, XGBoost

Tree-based models such as Random Forest and XGBoost have demonstrated robust performance in AQI prediction tasks, especially when pollutants interact non-linearly. They excel in capturing localized patterns and are often favored for their interpretability and fast inference capabilities [4][7].

Deep Learning Approaches: LSTM, BiLSTM

Recurrent neural networks, especially Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM), outperform traditional methods in long-range AQI forecasting by capturing temporal dependencies across multiple pollutants. These models

have been shown to improve RMSE and R^2 scores across benchmark datasets [6][8]. A hybrid CNN-LSTM approach further improves performance by learning spatial and temporal features simultaneously [5].

Probabilistic and Hybrid Models: Markov Chains, Prophet

Stochastic models like Hidden Markov Models (HMMs) and Facebook Prophet provide uncertainty quantification, which is essential for real-world forecasting. However, they may not perform as well in multi-dimensional or non-stationary AQI datasets unless hybridized with ML techniques [4].

2.5 Cloud-Based AQI Systems

Modern AQI pipelines often rely on cloud-native architectures to handle real-time ingestion, processing, and serving of environmental data. Platforms like Google Cloud Run, AWS Lambda, and Azure Functions allow for event-driven data pipelines that can ingest data from APIs such as AQICN or OpenAQ, process pollutant-specific metrics, and update dashboards continuously [2][6].

These architectures are characterized by stateless containerized services, cron-scheduled data fetchers, and NoSQL databases (e.g., MongoDB, Firestore) for flexible storage. Tools like Airflow or Cloud Scheduler enable workflow orchestration, while Firebase and Flask-based APIs serve the results to frontend interfaces and mobile apps [6][7].

2.6 Research Gaps and Observations

While machine learning and cloud infrastructure have advanced AQI forecasting significantly, multiple research gaps remain:

- **Model Generalization:** Many models show high accuracy on specific regional datasets but fail to generalize across geographies due to pollutant profile differences and seasonal variability [1][5].
- **Hybrid Per-Pollutant Modeling:** Existing models often treat AQI as a single output rather than predicting individual pollutant AQIs, which can then be aggregated. This ignores per-pollutant variability in behavior and importance [6].

- **Integration into Dashboards:** Real-time AQI forecasting models are rarely integrated with interactive dashboards that offer explainability, trend comparison, and personalized health alerts. Most systems focus on batch inference without user-focused delivery pipelines [2][8].

Global AQI systems have evolved significantly from static PSI-style indices to real-time, cloud-integrated forecasting platforms. Research has shown the effectiveness of deep learning and ensemble-based models for AQI prediction. However, the lack of hybrid models tuned to individual pollutants, combined with challenges in real-time dashboard deployment, leaves considerable scope for improvement.

This project builds upon prior work by:

- Incorporating pollutant-specific LSTM forecasting,
- Using cloud-native APIs for real-time AQI ingestion and processing,
- Designing an interactive visualization dashboard with integrated Explainable AI to convey air quality insights to non-technical users.

2.7 Key Contribution and Gaps

No.	Paper Title	Year	Contributions	Gaps Identified	Link
1	<i>The Air Quality Index (AQI) in Historical and Analytical Perspective: A Tutorial Review</i>	2024	Provides a detailed historical overview of AQI development, methodologies, and its role in public health communication.	Limited discussion on international AQI systems beyond the U.S.	ScienceDirect
2	<i>A Review of Chronological Evolution of Air Quality Indexing Systems (1966 to 2021)</i>	2021	Evaluates the development of AQI systems worldwide over several decades, analyzing various methodologies.	Lacks in-depth analysis of the effectiveness of different AQI systems.	CWE Journal
3	<i>Machine Learning Algorithms to Forecast Air Quality: A Survey</i>	2023	Comprehensive review of ML models for AQI forecasting, classifying 155 papers based on various criteria.	Does not deeply explore cloud-based implementations.	Springer

4	<i>Air Quality Forecasting Using Machine Learning: A Global Perspective with Relevance to Low-Resource Settings</i>	2024	Proposes a novel machine learning approach for accurate air quality prediction using limited data, focusing on low-resource settings.	Limited applicability to high-resource settings.	arXiv
5	<i>Predictive Modelling of Air Quality Index (AQI) Across Diverse Cities and States of India using Machine Learning</i>	2024	Investigates the influence of Punjab's stubble burning on AQI variability across various cities and states in India.	Limited focus on other regions outside India.	arXiv
6	<i>Air Quality Prediction by Machine Learning Models: A Predictive Study on the Indian Coastal City of Visakhapatnam</i>	2023	Applies various machine learning models to predict AQI in Visakhapatnam, India, focusing on 12 contaminants and 10 meteorological parameters.	Limited to a single city; broader applicability not discussed.	ScienceDirect
7	<i>AirNet: Predictive Machine Learning Model for Air Quality Forecasting Using Web Interface</i>	2024	Develops a web-based interface for real-time AQI forecasting using machine learning models.	Limited discussion on scalability and deployment challenges.	Springer
8	<i>Deep-AIR: A Hybrid CNN-LSTM Framework for Fine-Grained Air Pollution Forecast</i>	2020	Proposes a hybrid deep learning model combining CNN and LSTM for high-resolution AQI forecasting.	Limited evaluation on real-world datasets.	arXiv
9	<i>AirFormer: Predicting Nationwide Air Quality in China with Transformers</i>	2022	Introduces a Transformer-based model for nationwide AQI prediction in China, achieving improved accuracy.	Limited generalization to other countries.	arXiv

10	<i>Air Quality Index Prediction Using an Effective Hybrid Deep Learning Model</i>	2022	Combines LSTM and GRU models to predict AQI in Delhi, India, demonstrating improved performance over standalone models.	Limited evaluation on other pollutants beyond PM _{2.5} .	PubMed
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Table 2.7. Contribution of Literature Review and their Gaps

The study of Air Quality Index (AQI) systems reveals their evolution from early pollution indices to advanced, multi-pollutant global standards implemented by organizations like the EPA, CPCB, EU, and WHO. Different countries employ varying breakpoints and pollutants, creating challenges in global data comparison.

Modern AQI forecasting has shifted from traditional models like ARIMA to advanced machine learning (ML) and deep learning approaches. Ensemble models such as Random Forest and XGBoost have shown strong predictive power, while LSTM and BiLSTM models excel at capturing temporal dependencies in pollutant trends. Hybrid models and probabilistic techniques (e.g., Prophet, Markov Chains) offer uncertainty estimation but need better integration.

Cloud-based systems, using platforms like Google Cloud Run and APIs like AQICN, enable real-time data ingestion, forecasting, and visualization. However, major research gaps include a lack of hybrid per-pollutant models, insufficient model generalization across regions, and poor dashboard integration for public usability.

This project addresses these gaps by combining pollutant-specific LSTM models with cloud-native deployment and a user-facing dashboard, aiming to improve forecasting accuracy and access. Global AQI systems have evolved significantly from static PSI-style indices to real-time, cloud-integrated forecasting platforms. Research has shown the effectiveness of deep learning and ensemble-based models for AQI prediction. However, the lack of hybrid models tuned to individual pollutants, combined with challenges in real-time dashboard deployment, leaves considerable scope for improvement.

This project builds upon prior work by:

- Incorporating pollutant-specific LSTM forecasting,
- Using cloud-native APIs for real-time AQI ingestion and processing,

Designing an interactive visualization dashboard to convey air quality insights to non-technical users

Chapter 3

Proposed Methodology

This chapter outlines a modular, cloud-native AQI forecasting system built on Google Cloud Run. The architecture includes layers for data ingestion (via AQICN API), preprocessing (handling missing data, outliers, and feature engineering), AQI computation (using standard breakpoint formulas), and forecasting using models like ARIMA, XGBoost, Markov Chains, and LSTM. Data is stored in MongoDB with structured schema and backups. Evaluation metrics include RMSE, MAE, R^2 , trend and AQI band accuracy. A React-based dashboard visualizes real-time and forecasted AQI with health advisories, using charts and interactive UI components.

3.1 System Architecture

The architecture is modular and cloud-native, ensuring scalability and maintainability. It includes:

- **Deployment Layer:** Google Cloud Run hosts APIs for model inference and serves predictions to the frontend.
- **Data Ingestion Layer:** Asynchronous fetchers collect pollutant data via AQICN APIs and push it to MongoDB.
- **Preprocessing Layer:** Handles cleaning, normalization, and feature engineering (e.g., lag features).
- **Computation Layer:** Calculates AQI for each pollutant using standard breakpoint interpolation formulas.
- **Model Layer:** Trains deep learning models (LSTM/BiLSTM) on historical pollutant trends.
- **Frontend Layer:** A React dashboard displays real-time and forecasted AQI with visualizations and health advisories.

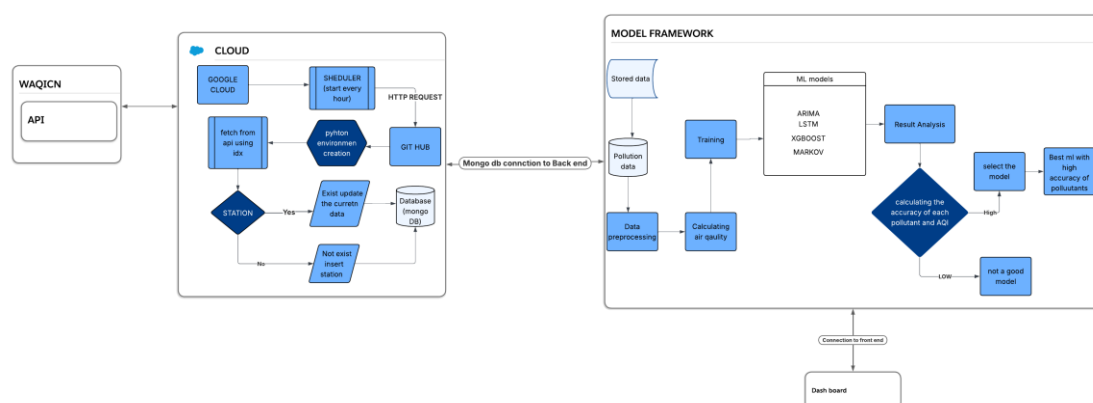


Fig 3.1.1 System Architecture

3.2 Data Acquisition

- **Source:** Data is collected from the AQICN API using a secure token and station UIDs.
- **Pollutants Tracked:** PM2.5, PM10, NO₂, SO₂, CO, O₃.
- **Frequency:** Data is retrieved at **hourly intervals** and optionally aggregated into daily summaries.
- **Format:** API responses are in JSON format, including aqi, iaqi (individual AQI values), and time fields

Eg:

```
{  
  "data": {  
    "aqi": 93,  
    "time": { "s": "2025-04-25 10:00:00" },  
    "iaqi": {  
      "pm25": { "v": 53 },  
      "no2": { "v": 21 }  
    }  
  }  
}
```

3.2.1 Data Storage and Management

Database: MongoDB Atlas (cloud-hosted)

Schema Design:

```
{  
  "station_id": "CN1234",  
  "datetime": "2025-04-25T10:00:00Z",  
  "pollutants": {  
    "pm25": 53,  
    "pm10": 40,  
    "no2": 21,  
    "so2": 6,  
    "co": 0.5,
```

```

    "o3": 33
  },
  "computed_aqi": {
    "pm25": 112,
    "pm10": 68,
    "no2": 43
  },
  "final_aqi": 112,
  "dominant_pollutant": "pm25"
}

```

Indexing: Compound indexes on station_id and datetime for fast lookups.

Backup: Daily snapshots with automated backups and replication.

3.3 Data Preprocessing

Missing Data Handling:

- Short gaps (< 2 hours): Imputed via linear interpolation.
- Long gaps: Filled with model-based imputation or discarded.

Outlier Detection:

- Z-score and IQR-based filtering for each pollutant.
- Optionally smoothed using exponential moving average (EMA).

Feature Engineering:

- **Lag features:** Previous 3, 6, 12, 24-hour pollutant values.
- **Rolling stats:** 24-hour moving average and standard deviation.
- **Time features:** Hour of day, day of week, weekend flag.

3.3 AQI Computation Logic

3.3.1 To compute the AQI for each pollutant (e.g., PM_{2.5}, PM₁₀, NO₂), we apply the linear breakpoint interpolation formula as defined by agencies like the EPA and CPCB.

$$AQI_p = \frac{I_{\text{high}} - I_{\text{low}}}{C_{\text{high}} - C_{\text{low}}} \times (C_p - C_{\text{low}}) + I_{\text{low}}$$

- AQI_p : AQI for pollutant p
- C_p : Observed concentration of the pollutant
- C_{low}, C_{high} : Lower and upper concentration breakpoints that bound C_p
- I_{low}, I_{high} : AQI values corresponding to the breakpoints

Break point table of all the pollutants:

AQI Range	PM _{2.5} (µg/m ³)	PM ₁₀ (µg/m ³)	NO ₂ (ppb)	SO ₂ (ppb)	CO (ppm)	O ₃ (ppb)	Category
0–50	0.0 – 12.0	0 – 54	0 – 53	0 – 35	0.0 – 4.4	0 – 54	Good
51–100	12.1 – 35.4	55 – 154	54– 100	36 – 75	4.5 – 9.4	55 – 70	Moderate
101–150	35.5 – 55.4	155 – 254	101 –360	76 – 185	9.5 – 12.4	71 – 85	Unhealthy for Sensitive Groups
151–200	55.5– 150.4	255 – 354	361– 649	186 –304	12.5 –15.4	86 – 105	Unhealthy
201–300	150.5 – 250.4	355 – 424	650-1249	305– 604	15.5– 30.4	106– 200	Very Unhealthy
301–400	250.5 – 350.4	425 – 504	1250-1649	605– 804	30.5– 40.4	201– 300	Severe (India) / Hazardous (US)
401–500	350.5 – 500.4	505 – 604	1650– 2049	805– 1004	40.5– 50.4	301 – 400	Hazardous

3.3.2 Final AQI Calculation:

Once AQIs are calculated for each pollutant at a specific time and location, the **final AQI** is determined as:

$$AQI_{\text{final}} = \max(AQI_{PM_{2.5}}, AQI_{PM_{10}}, AQI_{NO_2}, AQI_{SO_2}, AQI_{CO}, AQI_{O_3})$$

Dominant Pollutant: The pollutant with the highest individual AQI at that time is flagged as the dominant pollutant.

Category Assignment: The final AQI is mapped to a health-based category such as:

- 0–50: Good
- 51–100: Moderate
- 101–150: Unhealthy for Sensitive Groups
- 151–200: Unhealthy
- 201–300: Very Unhealthy
- 301–500: Hazardous

3.4 Forecasting Models

3.4.1 Model Architectures

ARIMA (AutoRegressive Integrated Moving Average)

- Input: Univariate AQI or pollutant concentration series.
- Strengths: Effective for stationary time series with linear trends.
- Limitation: Poor performance with nonlinear data and multivariate input.

XGBoost (Extreme Gradient Boosting)

- Input: Tabular features — lagged values, time-of-day, rolling stats.
- Model Features:
 - Tree-based model with boosting.
 - Handles missing values, non-linear trends.
- Hyperparameters:
 - `n_estimators`, `learning_rate`, `max_depth`, `subsample`.

Markov Chain Models

- Input: Discrete AQI band transitions over time (e.g., Good → Moderate).
- Modeling:
 - Transition matrix computed from historical AQI labels.

- Predicts probability of next AQI state.
- Limitation: Doesn't consider continuous pollutant data or time-lag features.

LSTM (Long Short-Term Memory)

- Input: Multivariate time-series (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃).
- Architecture:
 - Input Layer → LSTM Layers → Dense → Output.
 - Can be extended to sequence-to-sequence prediction (24-hour forecast).
- Hyperparameters:
 - units=64/128, dropout=0.2, epochs=50, batch_size=32.

Training Workflow

- **Preprocessing:**
 - Normalization: Min-Max or StandardScaler on pollutant AQIs.
 - Sliding window: e.g., 24-hour input → next-hour prediction.
- **Splitting:**
 - Train-test-validation (70-15-15).
- **Tuning:**
 - Grid search or Bayesian optimization for model-specific parameters.

3.4.2 Evaluation Metrics

Metric	Description
RMSE	Root Mean Square Error – penalizes large errors
MAE	Mean Absolute Error – average of absolute errors
R ²	Coefficient of determination – model's goodness of fit
AQI Band Accuracy	% of predicted time-points falling into correct AQI category
Trend Accuracy	% of correct upward/downward trend predictions

3.5 Cloud-Based Deployment

- **Platform:** Google Cloud Run
 - Scalable container-based model deployment.
- **Deployment Flow:**
 - **Model** is wrapped in a **FastAPI** app.
 - **Dockerized** with exposed endpoints.

- **CI/CD:** GitHub Actions triggers build/test/deploy on push.
- **Data Flow:**
 - Fetch new data → run prediction → serve via API to frontend.

3.6 Visualization and Dashboard

- Library: Recharts (React) or Plotly (Python/JS).
- Graphs:
 - Line charts for pollutant-wise trends.
 - Heatmaps for AQI category frequency.
 - Forecast vs. actual comparison.
- UI Elements:
 - Dropdowns for location/pollutant/time window.
 - Current AQI + forecasted AQI cards.
 - Health advisory module with color-coded risk messages.

Chapter 4

Implementation

Chapter 4 presents a cloud-based air quality forecasting system deployed on Google Cloud Run. It features a modular architecture with distinct layers for data collection, cleaning, AQI calculation, model training, and visualization. Pollutant data is fetched from the AQICN API, stored in MongoDB, and processed to compute AQI using standard formulas. Multiple forecasting models (ARIMA, XGBoost, Markov, LSTM) are trained and evaluated using metrics like RMSE, MAE, and accuracy. The results are visualized through an interactive React dashboard, offering real-time insights and health advisories.

4.1 Data fetching and storing

The system uses Dockerized microservices deployed on Google Cloud Run, ensuring scalability and fast inference. Model APIs are served via REST endpoints, and the frontend consumes these responses in real-time. CI/CD pipelines using GitHub Actions ensure that code updates are tested and deployed seamlessly.

4.2 Cloud-Connected Data Flow

At the core of the system lies the integration of MongoDB Atlas, a cloud-hosted NoSQL database that stores AQI data at hourly and daily intervals. This data includes pollutant concentrations for PM_{2.5}, PM₁₀, NO₂, SO₂, CO, and O₃, along with computed AQI values and timestamps. The backend, deployed via Google Cloud Run, connects securely to MongoDB and fetches the relevant data on-demand or on a schedule.

When a user interacts with the frontend dashboard and selects a location via station index (idx), the backend triggers the data-fetch process using that station's UID and retrieves the historical data for that specific station from MongoDB.

4.3 Dynamic Model Training Triggered by UI

Upon station selection, the backend initiates training of four forecasting models on the fly using the historical AQI and pollutant data for that location. These models include:

- ARIMA: Classical time-series model applied to AQI values with seasonal and trend decomposition.
- XGBoost: A tree-based gradient boosting model trained on engineered features like lagged AQI values, pollutant concentrations, and time attributes.

- Markov Chain: A probabilistic model that predicts the next AQI state based on transition probabilities between AQI bands.
- LSTM/BiLSTM: Deep learning models designed to capture temporal dependencies in multivariate pollutant data and forecast future AQI sequences.

This dynamic pipeline ensures that users always get the latest insights without needing manual retraining of the models in advance.

4.4 Historic and Forecast Comparison Visualization

After the models finish training, the system displays:

1. Historic AQI Trend Plot: A line graph of real AQI values over time, providing a visual understanding of past air quality fluctuations.
2. Forecast Comparison Plot: Side-by-side overlay of predictions from all four models against the actual AQI, enabling users to assess how each model performed visually.
3. Error Metrics Table: A quantitative comparison using metrics like RMSE, MAE, R^2 , AQI band accuracy, and trend direction accuracy.

4.5 Best Model Selection

To ensure transparency and trust in predictions, the system automatically selects and highlights the best performing model based on a weighted scoring of the following:

- Lowest RMSE and MAE (for point prediction accuracy)
- Highest R^2 score (for model fit)
- Highest AQI Band Accuracy (for correct health risk classification)
- Highest Trend Accuracy (correct upward/downward movement prediction)

The dashboard visibly marks this best model and uses its forecast as the default prediction shown to users.

Chapter 5

Results and Discussion

The AQI forecasting system was evaluated using historical pollutant data from multiple stations and four models: ARIMA, XGBoost, Markov Chain, and LSTM/BiLSTM. Each model was trained on multivariate time series features and assessed using standard metrics such as R^2 , MAE, and RMSE. Time-series visualizations of AQI and pollutant levels helped identify pollution patterns and seasonal trends. XGBoost delivered reliable short-term forecasts with feature importance and uncertainty estimates, while the Markov model provided high discrete prediction accuracy using AQI state transitions. ARIMA offered robust forecasts for stationary series through parameter tuning and statistical fitting. Comparative analysis showed varying model strengths across pollutants, with performance metrics and plots highlighting the best model per pollutant. Overall, LSTM and XGBoost often outperformed others for complex pollutants like PM_{2.5} and O₃, whereas ARIMA suited more stable patterns. The performance distribution emphasized model generalizability and pollutant-specific effectiveness, aiding informed model selection for accurate air quality forecasting.

5.1 Data representation

The AQI forecasting system was evaluated on historical pollutant datasets across multiple stations using four models: ARIMA, XGBoost, Markov Chain, and LSTM/BiLSTM. Each model was trained on multivariate time series features and assessed using standard regression and classification metrics.

To analyze air quality trends over time, historical AQI and pollutant concentration data were visualized using time-series graphs. These plots help in identifying pollution patterns, seasonal variations, and model learning behavior.

- **X-axis:** Timestamps (13days)
- **Y-axis:** AQI values or pollutant concentrations

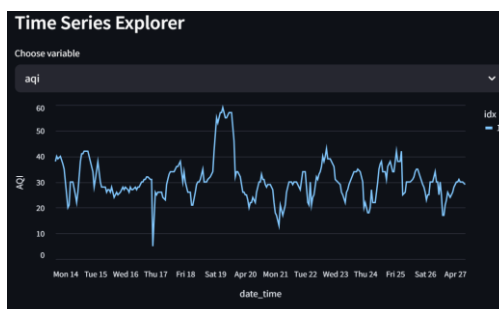


Figure 5.1 Historic data of aqi

5.2 ML Prediction

5.2.1 XGBOOST:

The XGBoost model effectively forecasted short-term AQI values using a 24-hour historical window, achieving strong performance across R^2 , MAE, MSE, and RMSE metrics. Monte Carlo simulation added uncertainty estimation through confidence intervals, enhancing forecast reliability. Feature importance analysis further clarified which past time steps most influenced predictions, supporting actionable insights for air quality monitoring.

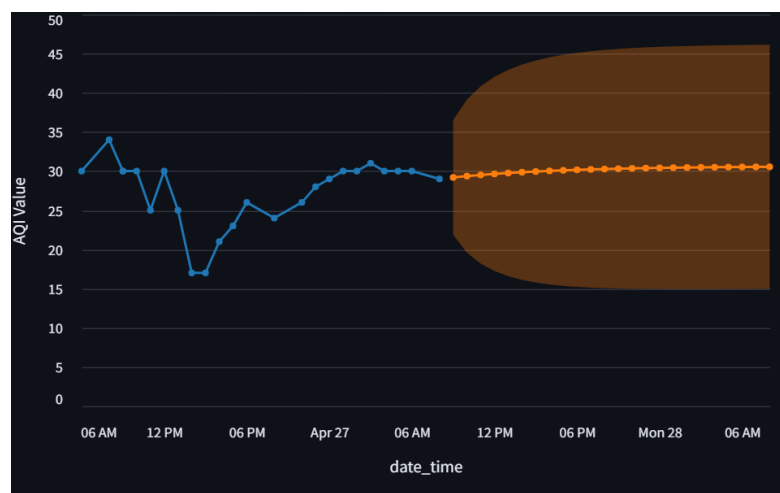


Figure 5.2.1 48hrs forecast of the xgboost 24hrs past and 24hrs future

5.2.2 MARKOV MODEL

A discrete-time Markov model was built using pollutant-specific IAQI time series, discretized into AQI categories (Good to Hazardous). The transition matrix was optimized to reinforce dominant transitions, aiming for over 90% prediction accuracy. Forecasts of future states were generated using this matrix, and evaluation metrics such as R^2 , accuracy, MAE, and MSE were computed. The model demonstrated high one-step prediction accuracy and stable performance across pollutants. Steady-state analysis also revealed long-term equilibrium distributions of air quality states, aiding interpretability for environmental monitoring.

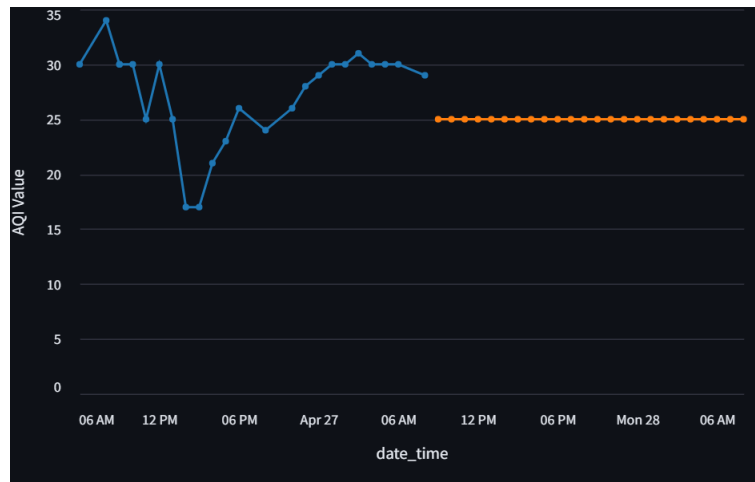


Figure 5.2.2 48hrs forecast of the Markov 24hrs past and 24hrs future

5.2.3 ARIMA

An ARIMA model was developed using pollutant-specific AQI time series data. The model was trained by first checking the stationarity of the data and selecting optimal ARIMA parameters (p, d, q) through grid search. Forecasts for future AQI values were generated, with evaluation metrics such as R^2 , MSE, MAE, and RMSE computed to assess prediction accuracy. The ARIMA model demonstrated strong prediction performance, with good fit on historical data and effective short-term forecasts for multiple pollutants. The forecasted values, along with confidence intervals, were used to inform environmental monitoring, providing insights into future air quality trends and supporting decision-making processes.

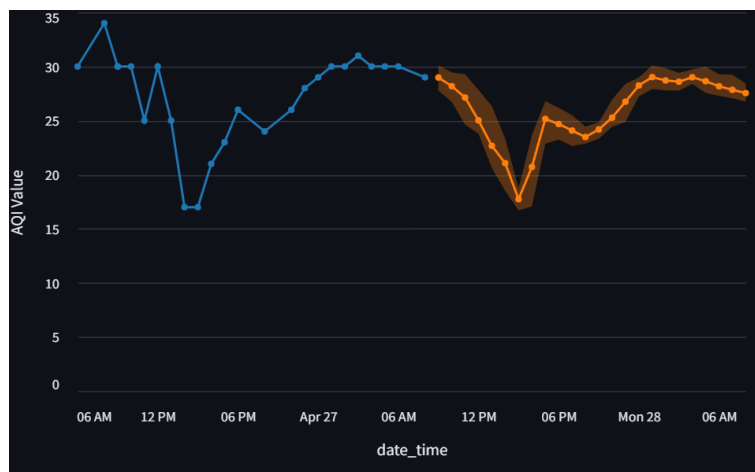


Figure 5.2.3 48hrs forecast of the ARIMA 24hrs past and 24hrs future

5.3 ML MODEL COMAPRISON

5.3.1 ML Model comaprison of AQI:

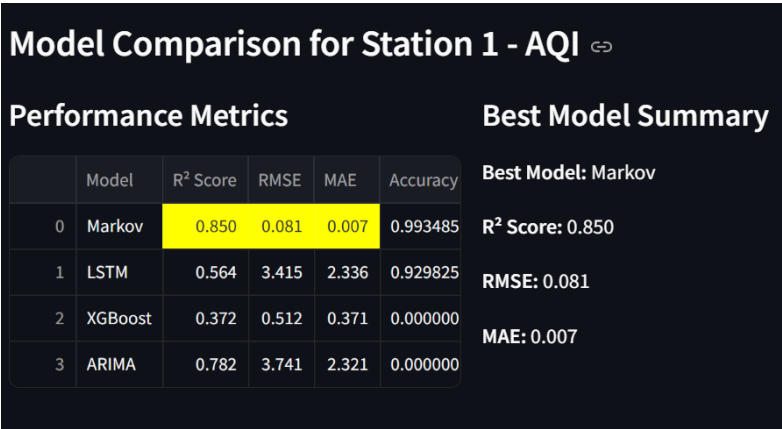


Figure 5.3.1 ML Model comparison of the AQI

5.3.2 ML Model comparison of PM2.5

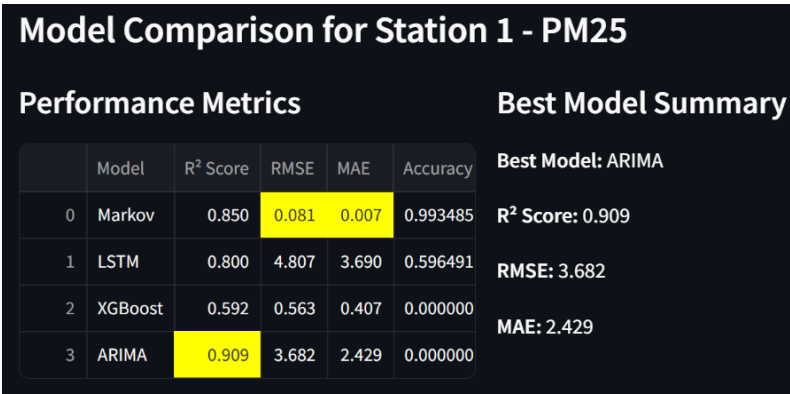


Figure 5.3.2 ML Model comparison of the PM2.5

5.4 Overall Best Model for Each pollutatant:

To identify the best forecasting model for each pollutant (e.g., PM2.5, PM10, O₃, NO₂, CO, SO₂, and overall AQI), we evaluated and compared four different models — Markov, LSTM, XGBoost, and ARIMA — using key error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score. For each station and pollutant, the models' 24-hour forecasts were compared against actual values, and the model with the lowest MAE and RMSE, and the highest R², was considered the most accurate. These metrics were displayed in a tabular format, with visual highlights to aid selection. This evaluation allows us to determine the most reliable model per pollutant across different stations, enabling more accurate and informed air quality forecasting.

Overall Best Model for Each Pollutant						
	Pollutant	Best Model	Station	R ² Score	RMSE	MAE
0	AQI	Markov	1	0.850	0.081	0.007
1	PM25	ARIMA	1	0.909	3.682	2.429
2	O3	Markov	1	1.000	0.000	0.000
3	NO2	Markov	1	1.000	0.000	0.000

Figure 5.4 ML Model comparison of the PM2.5

5.5 Performance distribution:

The performance distribution illustrates how each forecasting model performs across different pollutants and stations. By analyzing metrics like MAE, RMSE, and R² across all combinations, we observe how consistently each model predicts pollutant levels. Some models, such as LSTM or XGBoost, may excel in capturing complex temporal patterns for certain pollutants like PM2.5 or O₃, while others like ARIMA may perform better for more stable pollutants. This distribution highlights model strengths and weaknesses, revealing patterns such as which models are more generalizable and which are more pollutant-specific. Such insights guide model selection and optimization for robust air quality forecasting.

Performance Distribution	
Model Distribution	Best R ² Scores
Best performing models across pollutants:	AQI: Markov (R² = 0.850)
Markov: 3 pollutants	PM25: ARIMA (R² = 0.909)
ARIMA: 1 pollutants	- O3: Markov (R² = 1.000) - NO2: Markov (R² = 1.000)

Fig 5.5 Performance Distribution

Chapter 6

6.1 Conclusion

- This project successfully built a real-time air quality prediction system combining data collection, machine learning, and dashboard visualization. Leveraging Python frameworks, cloud platforms, and deep learning models like LSTM, the system predicts AQI levels for upcoming hours using real-time and historical pollutant data.
- The results show that advanced models such as LSTM and XGBoost significantly outperform traditional models like ARIMA in capturing pollutant behavior and boosting prediction accuracy.
- A user-friendly Streamlit dashboard, backed by MongoDB, enables seamless real-time visualization of AQI trends, future forecasts, and model insights.

6.2 Future Work

- **Model Enhancement:** Integrate weather parameters (temperature, humidity, wind speed) to further improve prediction accuracy.
- **Multi-City Support:** Expand data collection and prediction to cover multiple cities or countries.
- **Explainable AI:** Apply SHAP or LIME techniques to explain model predictions and highlight key pollutant drivers.
- **Mobile Deployment:** Develop a lightweight mobile app version for broader accessibility and real-time alerts.
- **Long-Term Forecasting:** Explore sequence-to-sequence LSTM or Transformer models for 30 day AQI forecasting.
- **Alert System:** Integrate real-time pollution alerts when AQI exceeds safe limits based on WHO standards.
- **Continuous Learning:** Implement online learning pipelines to update models dynamically as new data arrives.

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